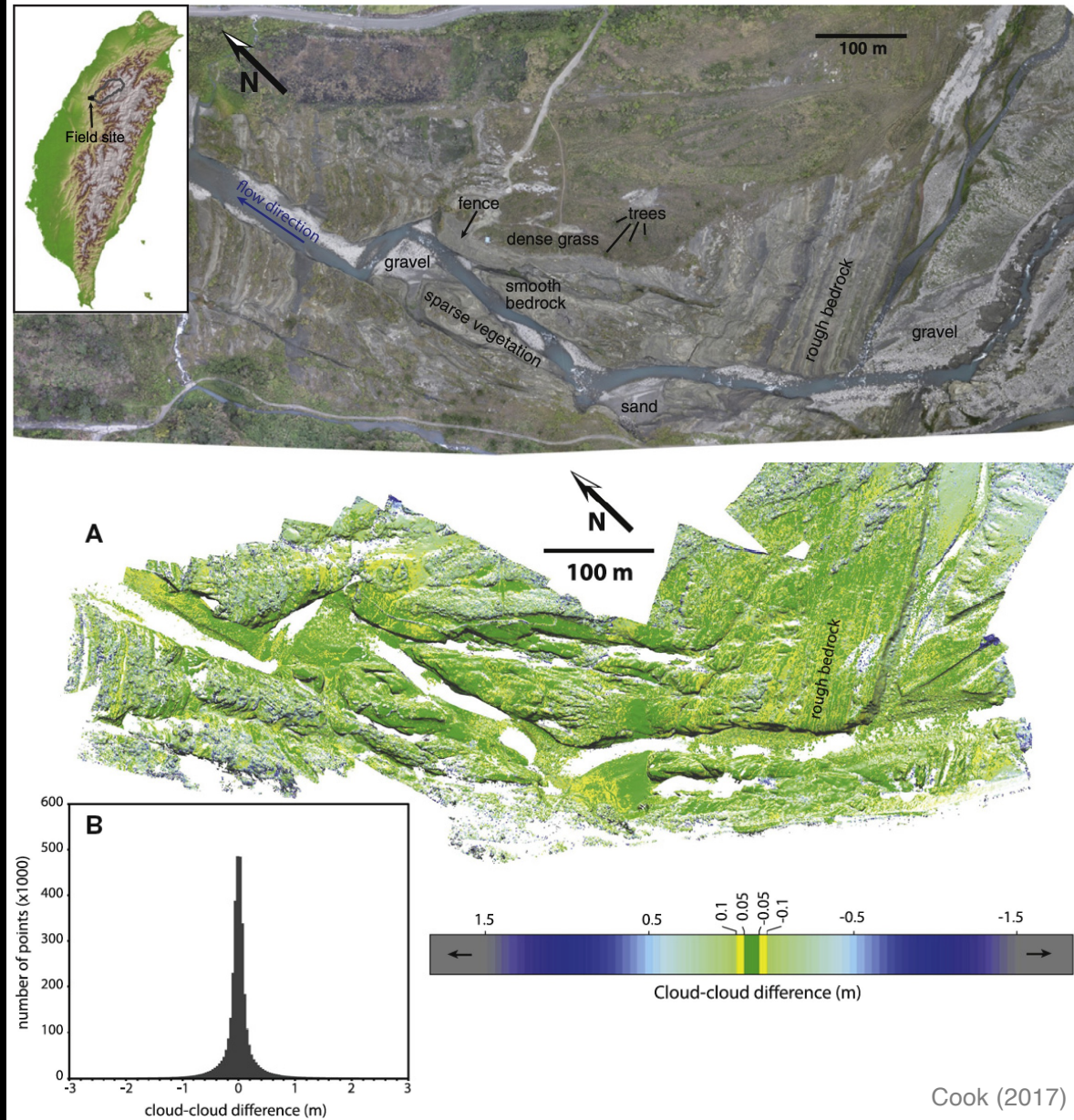


Terrain Analysis

GEOG 1026
Department of Geography
National Taiwan University



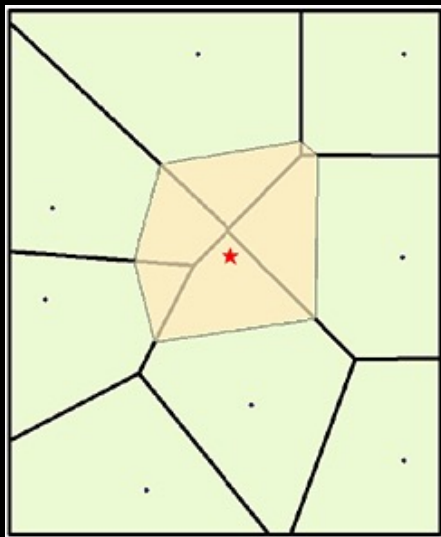
Cook (2017)

Fundamental Geo-Processing and GIS Analysis

- Spatial and attribute query
- Data transformation
- Data management
- Spatial extraction
- Spatial overlay
- Proximity analysis
- Spatial statistics analysis
- Data classification
- Spatial interpolation
- Terrain analysis
- Network analysis

Natural Neighbor (自然鄰域法)

- The method finds the closet subset of input samples to a query point and applies weights to them based on proportionate areas to interpolate a value.
- It is also known as Sibson or “area-stealing” interpolation.
- The proportion of overlap between the new polygon (beige) created around the interpolation point (red star) and the initial polygons (olive) is then used as the weights.

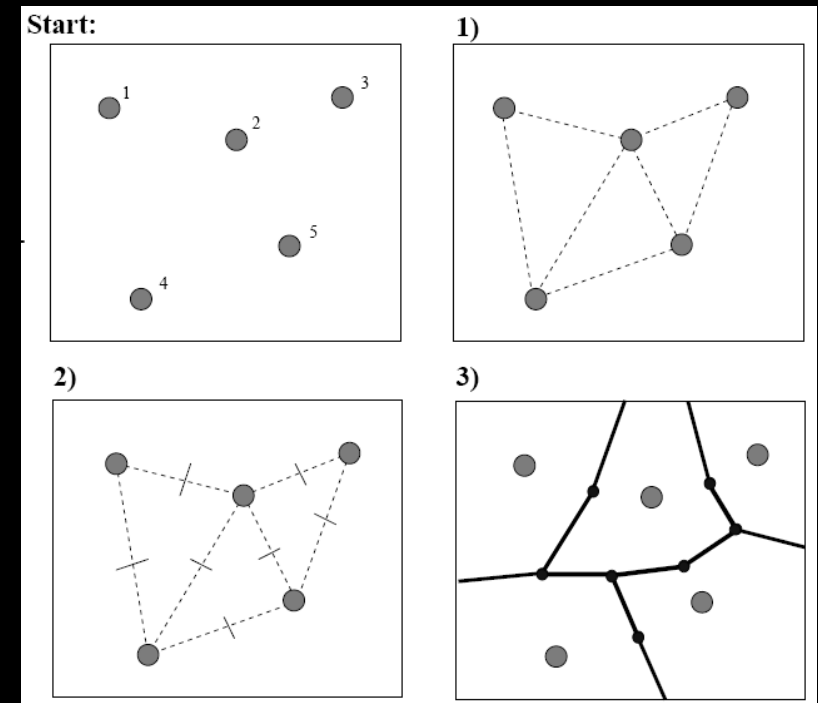


$$u(x) = \sum_{i=1}^n w_i u(x_i) \quad w_i = \frac{A(x_i)}{A(x)}$$

- $A(x_i)$ is the volume of intersection between the new polygon centered in x and the old polygon centered in x_i .
- $A(x)$ is the volume of the new polygon centered in x .

Voronoi Diagrams

- Also called Voronoi tessellation, Thiessen polygons, Dirichlet tessellation, etc.
- Points are triangulated into Delaunay triangulation (or TIN in GIS).
- The perpendicular bisectors for each triangle edge are constructed to form the edges of the Voronoi diagrams.
- The intersections become the polygon vertices.
- The Input Coverage point locations become label point locations.



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Purpose of Terrain Analysis

- Deriving basic terrain properties
- Providing visualization for geomorphological interpretation
- Environmental analysis and modeling

Spatial Data for Geography

Land cover change

Military unit detection
Land use change
Disaster coverage
Agriculture and forestry
Ships and oil spills

Topographic change

Topographic analysis
Landslide volume
Mining volume
Building structure
Forester structure

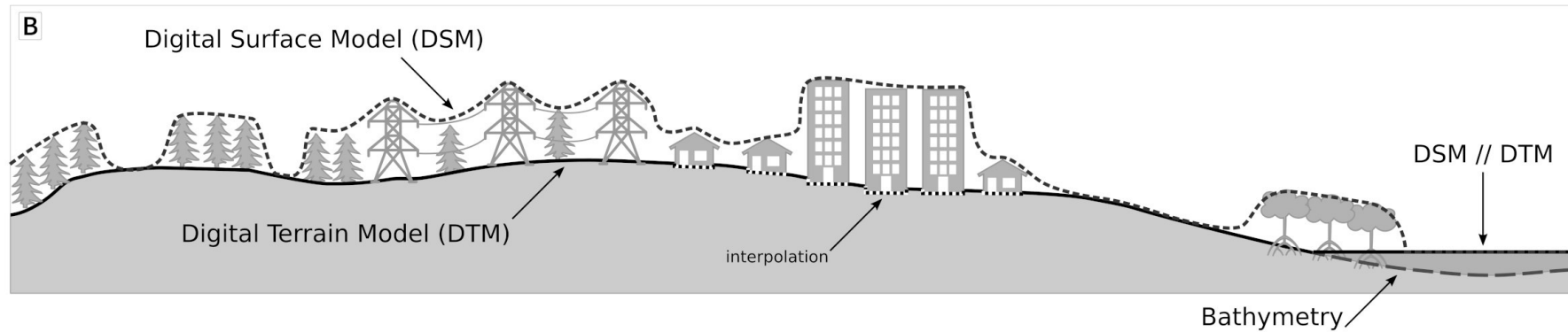
Surface deformation

Earthquakes
Subsidence
Surface creep
Volcanoes
Infrastructure



Elevation Models

- Digital Elevation Model (DEM)
 - Narrow sense: Elevation of bare earth
 - Broad sense: Elevation of surface
- Digital Terrain Model (DTM)
 - In US: DEM with terrain features
 - Other places: = DEM (narrow sense)
- Digital Surface Model (DSM)
 - Top of everything on the surface
- Digital Height Model (DHM)
 - $DSM - DEM$ (narrow sense)
- Non-Vegetated Surface (NVS)
- Canopy Height Model (CHM)

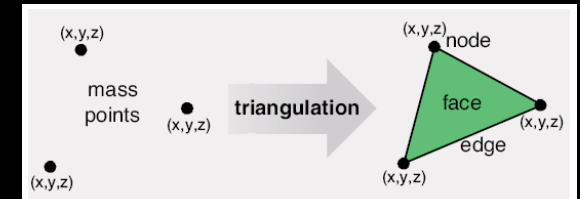


Representation of Surface in GIS

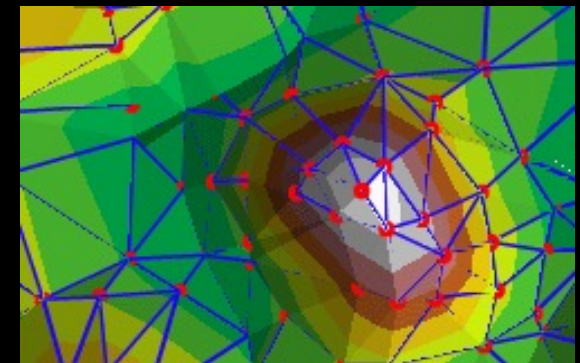
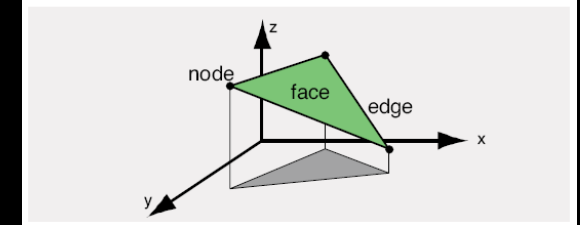
- Elevation models in raster format
- Triangulated Irregular Network (TIN)
- Point Cloud
- Contour

Triangulated Irregular Network, TIN (不規則三角格網)

- TIN is a vector-based data format.
- Represent surfaces as contiguous non-overlapping triangular faces.
 - The TIN representation models a surface from a set of points from which triangles are formed, or **triangulated**.
 - Triangles are made from three points that occur at **irregular** locations.
 - Each triangle stores topological information about its neighboring triangles, thus forming a **network**.
- Preserves the precise location and shape of surface features.
- Support a variety of surface analysis
- Well suited for large-scale mapping applications where positional accuracy and shapes of surface features are important.

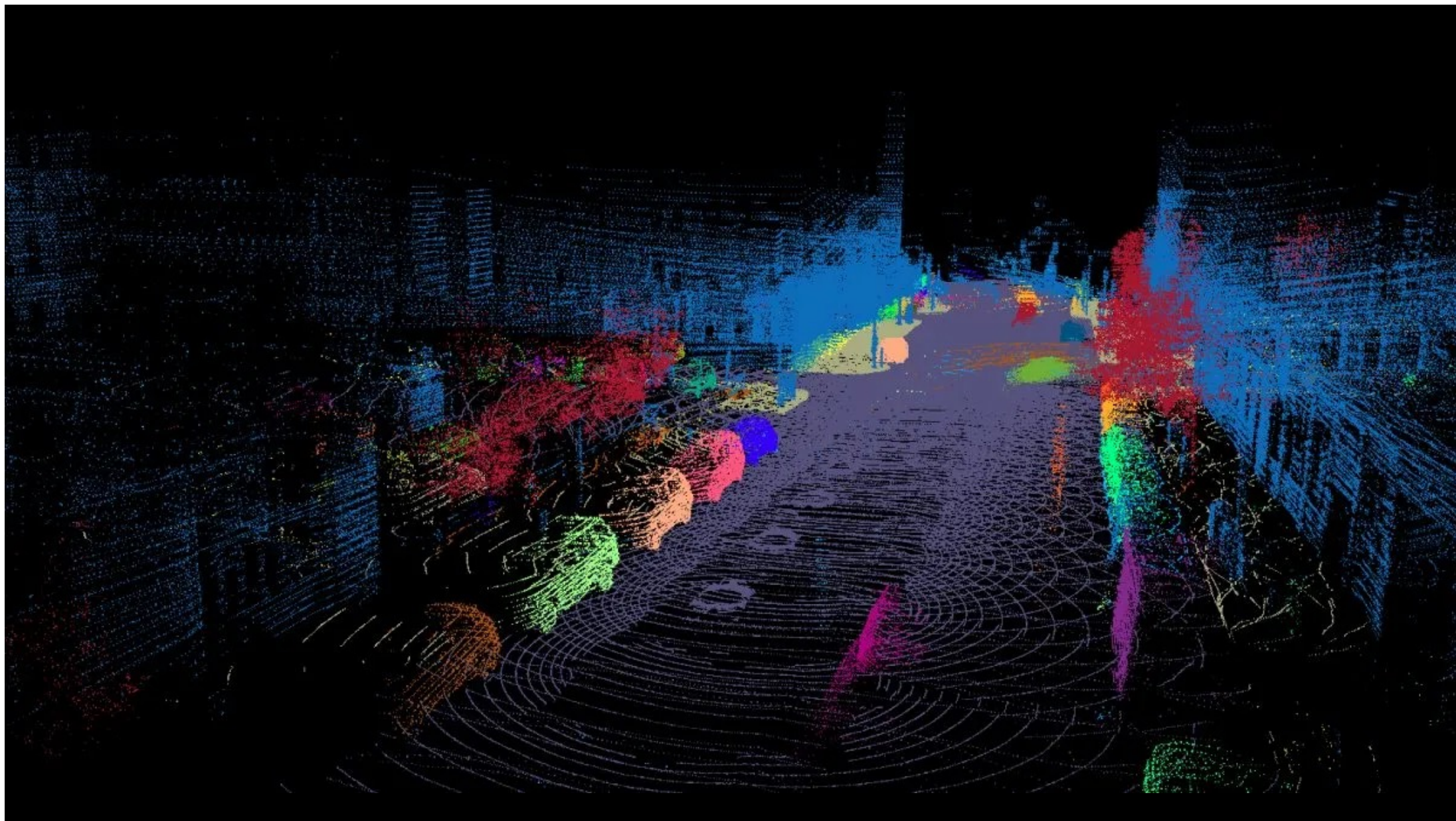


Each face in a TIN is a part of a plane in three-dimensional space. All of the faces in a TIN meet their neighbors precisely at each node and along each edge. Faces cannot intersect each other.



Point Cloud (點雲)

- A set of discrete points with 3D coordinates
- Commonly from LiDAR (Light Detection and Ranging) scan, SfM (Structure from Motion), and photogrammetry
- Point clouds may also have attributes of intensity, color, classification, etc.
- Usually the data sources of elevation models

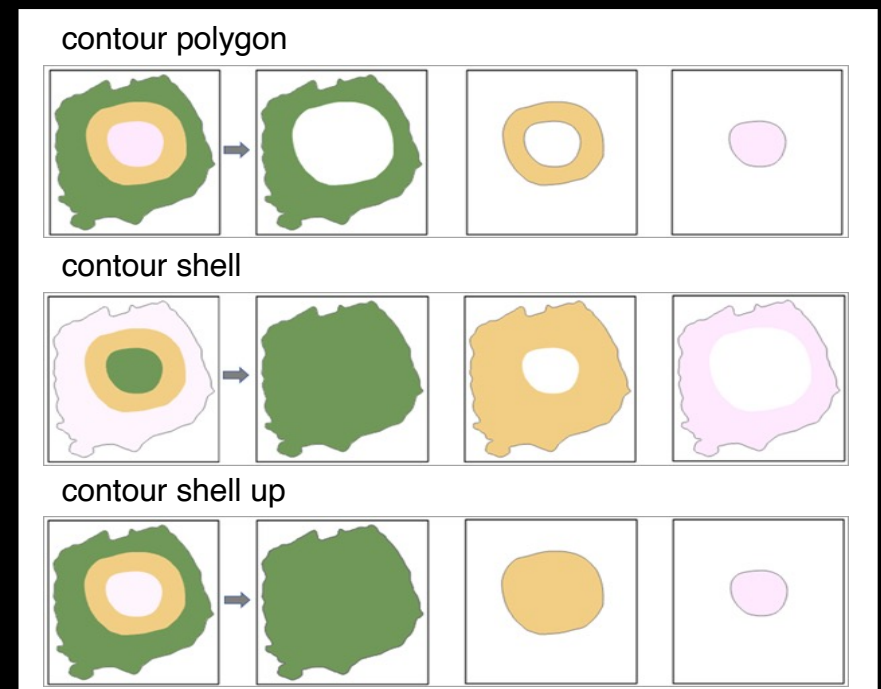


Common Terrain Analysis and Surface Toolsets

- Contour (等高線)
- Aspect (坡向)
- Slope (坡度)
- Curvature (曲率)
- Hillshade (地形陰影、暈渲)
- Viewshed analysis (視域分析)

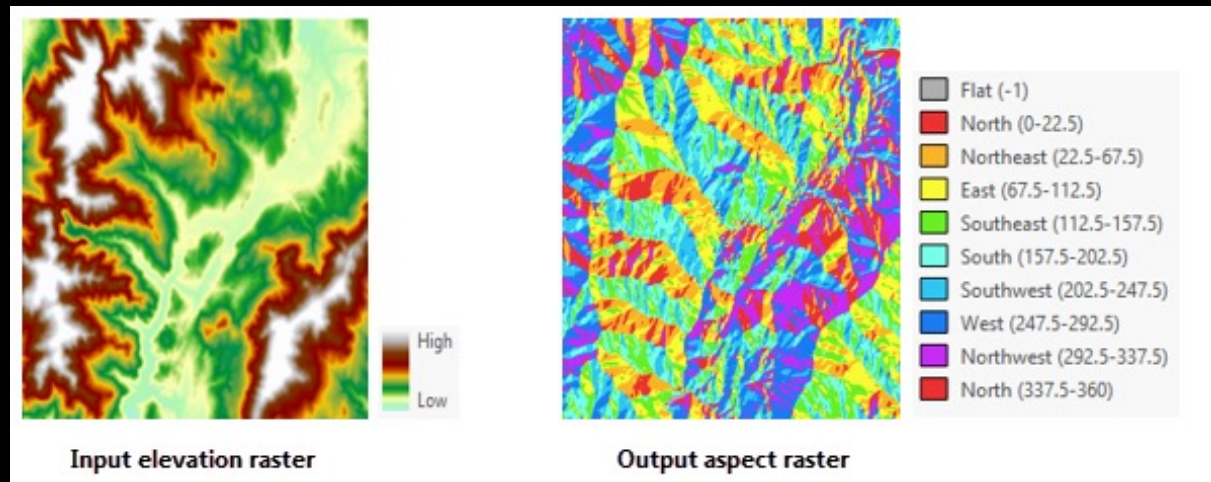
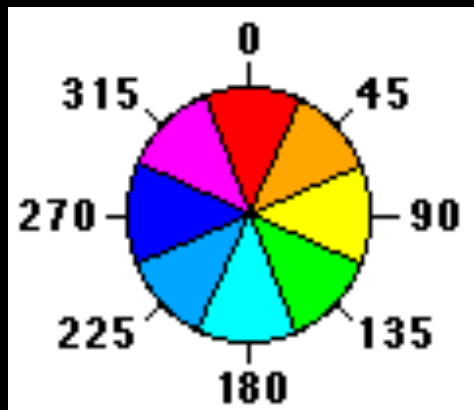
Contour (等高線)

- In ArcGIS Pro, contour lines are calculated from raster data by using linear interpolation to trace pathway of equal values across raster cells.
- Contour type:
 - In example of a raster with values between 0 and 575 and contour interval of 250.
 - Contour: Lines at 250 and 500
 - Contour polygon: Nonoverlapping polygons between 0-250, 250-500, and 500-575.
 - Contour shell: Overlapping polygons between 0-575, 0-500, and 0-250.
 - Contour shell up: Overlapping polygons between 0-575, 250-575, and 500-575.

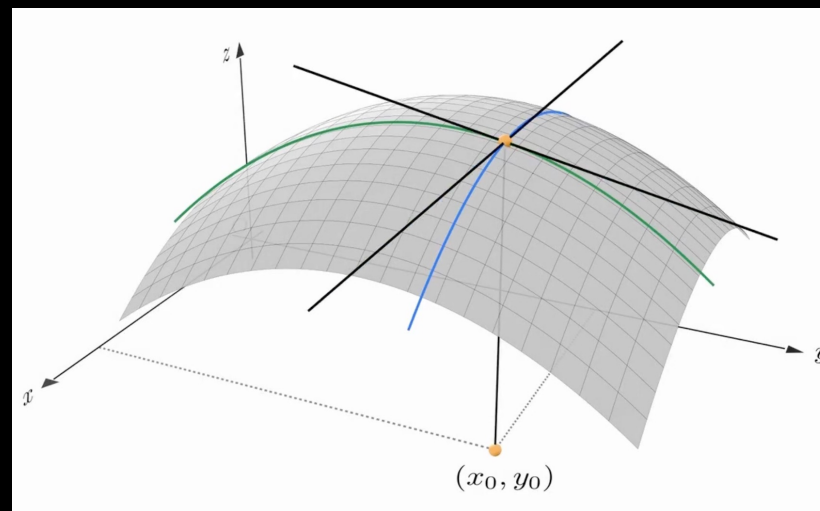
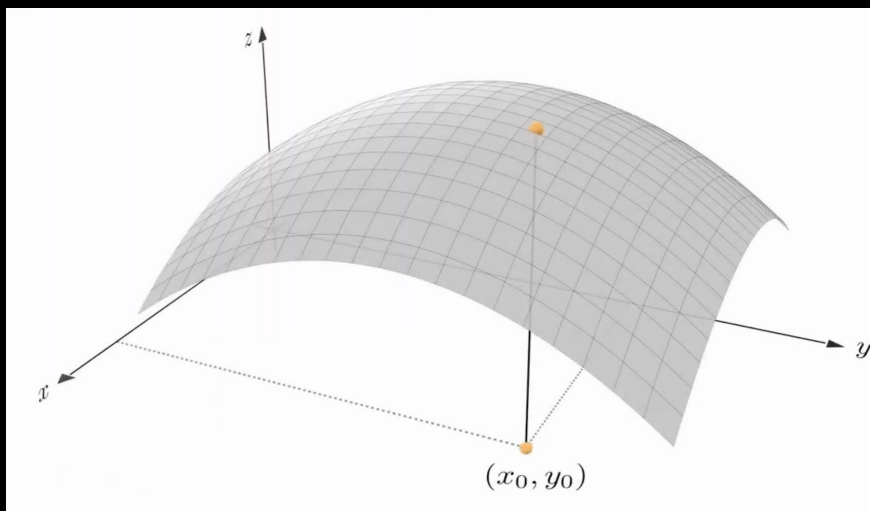


Aspect (坡向)

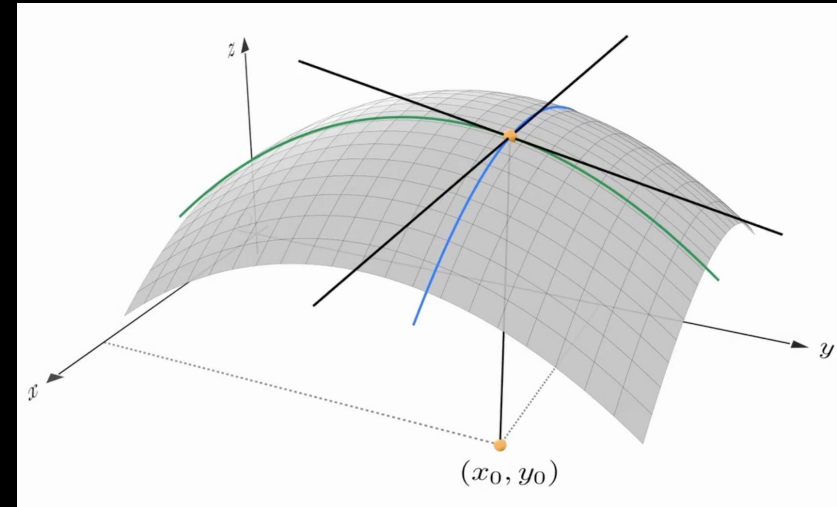
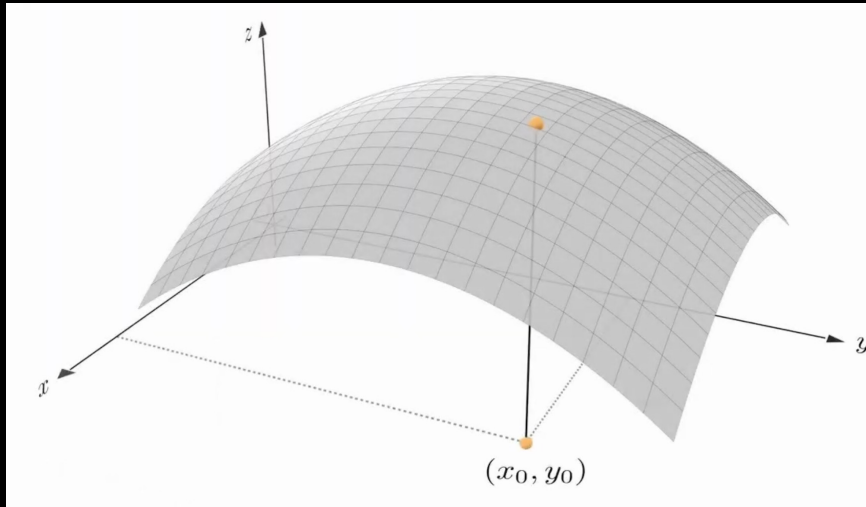
- The value of the aspect indicate the compass direction (measured clockwise from 0° to 360°) the surface faces at the location.
- The flat areas having no downslope direction are given a value of -1 in ArcGIS pro.



Slope (坡度)



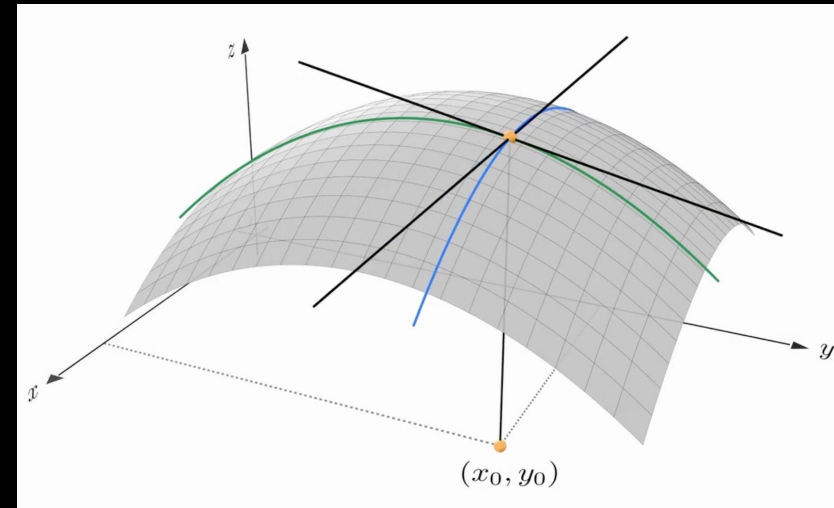
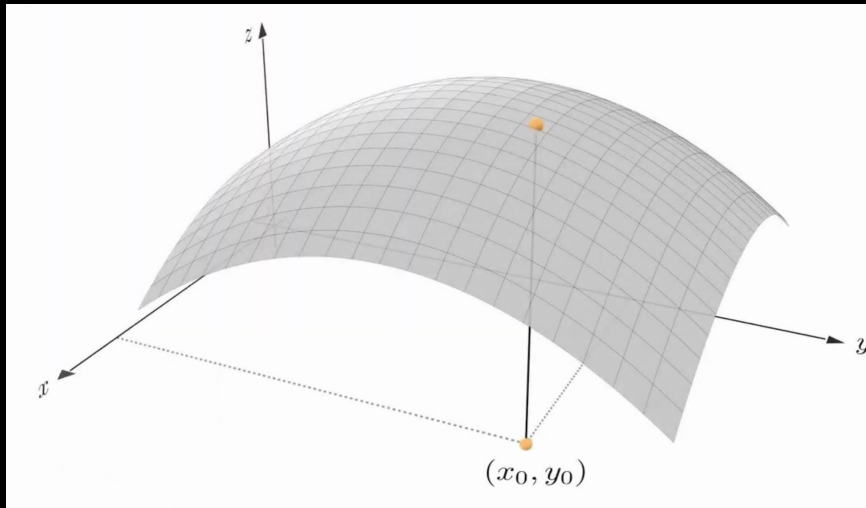
Slope (坡度)



- Height is $z = f(x, y)$, and slope (rate of change) in x direction is $\frac{\partial z}{\partial x}$ and in y is $\frac{\partial z}{\partial y}$
- At location (a, b) , $\frac{\partial f(a, b)}{\partial x} = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$ and $\frac{\partial f(a, b)}{\partial y} = \lim_{k \rightarrow 0} \frac{f(a, b + k) - f(a, b)}{k}$
- Ex. $z = f(x, y) = 2x^2 + y^2$

$$\frac{\partial z}{\partial x} = ? \quad \frac{\partial z}{\partial y} = ?$$

Slope (坡度)



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- Slope in the direction of unit vector (單位向量) $u \langle m, n \rangle$:

$$D_u f(x, y) = \nabla f(x, y) \cdot u = \frac{\partial z}{\partial x} \cdot m + \frac{\partial z}{\partial y} \cdot n \quad \nabla f(x, y) = \left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y} \right)$$

Dot product (內積)
Nabla (梯度)

$$u = \frac{v}{|v|} = \frac{\langle x_0, y_0 \rangle}{\sqrt{x_0^2 + y_0^2}} = \langle m, n \rangle,$$

$m^2 + n^2 = 1$

Slope (坡度)

- In ArcGIS, slope at cell *e* is interpreted as:

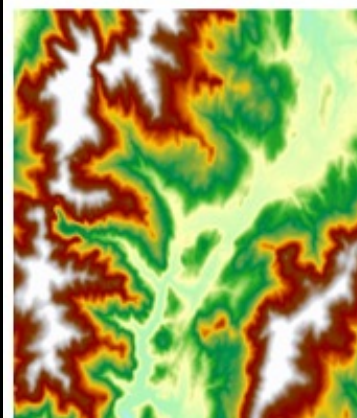
$$\text{Slope} = \arctan \left(\sqrt{\frac{dz^2}{dx} + \frac{dz^2}{dy}} \right)$$

$$\frac{dz}{dx} = \frac{\left((c + 2f + i) * \frac{4}{wgh1} - (a + 2d + g) * \frac{4}{wgh2} \right)}{8 * x_cellsize}$$

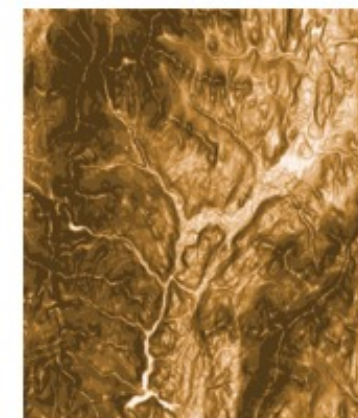
$$\frac{dz}{dy} = \frac{\left((g + 2h + i) * \frac{4}{wgh3} - (a + 2b + c) * \frac{4}{wgh4} \right)}{8 * y_cellsize}$$

wgh1 ... *wgh4* are weighted counts in valid cell

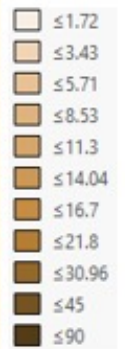
a	b	c
d	e	f
g	h	i



Input elevation raster

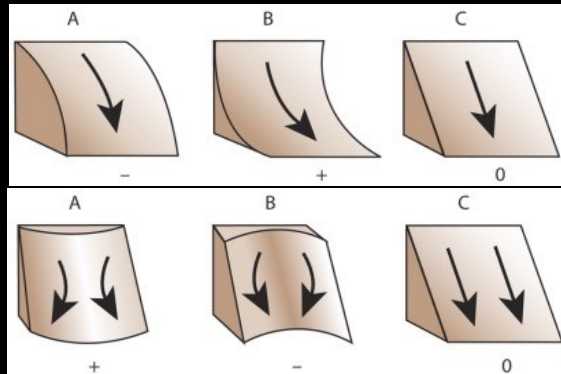


Output slope raster
(in degrees)



Curvature (曲率)

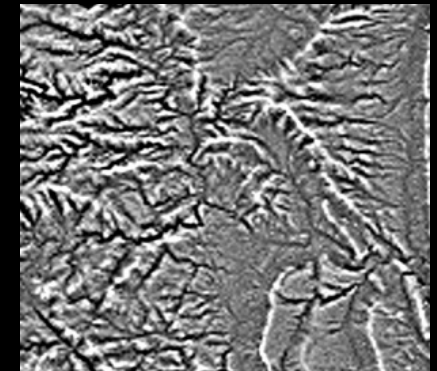
- Height is $z = f(x, y)$, and curvature is the rate of slope change $f''(x, y)$.



Profile curvature (垂直曲率): $\frac{d^2 z}{dy^2}$

Plain curvature (平面曲率): $\frac{d^2 z}{dx^2}$

Standard curvature = $\left(\frac{d^2 z}{dx^2} + \frac{d^2 z}{dy^2} \right)$



- In ArcGIS pro, a fourth-order polynomial is applied to each cell:

$$Z = Ax^2y^2 + Bx^2y + Cxy^2 + Dx^2 + Ey^2 + Fxy + Gx + Hy + I$$

$$A = ((Z1 + Z3 + Z7 + Z9)/4 - (Z2 + Z4 + Z6 + Z8)/2 + Z5)/L^4$$

$$B = ((Z1 + Z3 - Z7 - Z9)/4 - (Z2 - Z8)/2)/L^3$$

$$C = ((-Z1 + Z3 - Z7 + Z9)/4 + (Z4 - Z6)/2)/L^3$$

$$D = ((Z4 + Z6)/2 - Z5)/L^2$$

$$E = ((Z2 + Z8)/2 - Z5)/L^2$$

$$F = (-Z1 + Z3 + Z7 - Z9)/4L^2$$

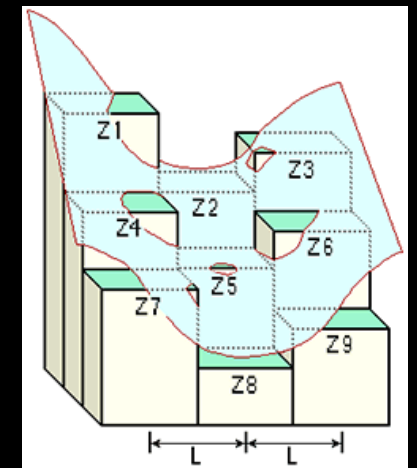
$$G = (-Z4 + Z6)/2L$$

$$H = (Z2 - Z8)/2L$$

$$I = Z5$$

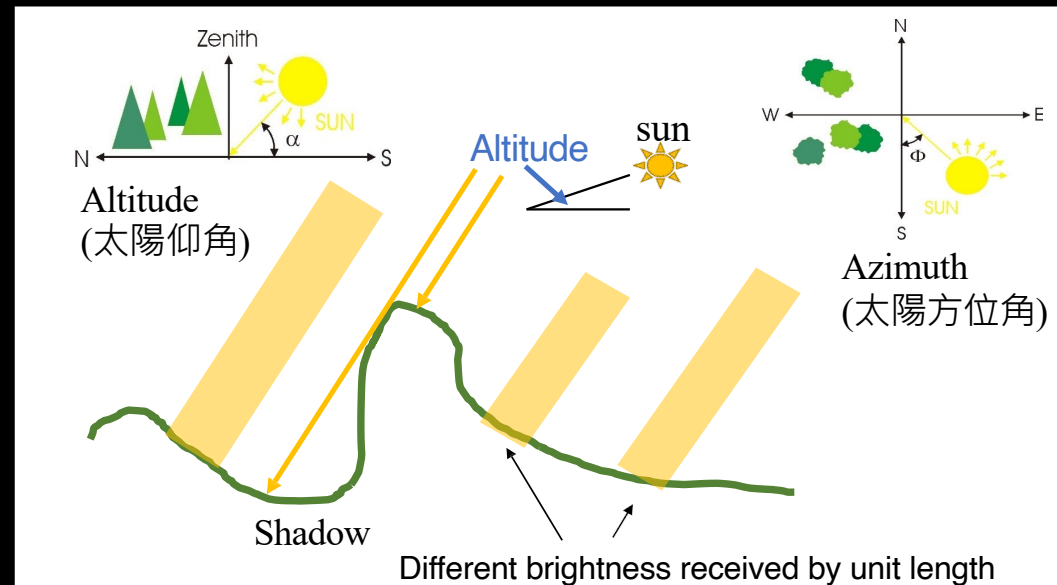
$$\text{Standard curvature} = -100 * \left(\frac{d^2 z}{dx^2} + \frac{d^2 z}{dy^2} \right)$$

For visualization



Hillshade (地形陰影、暈渲)

- In GIS, hillshading is a technique used to simulate how light interacts with terrain to create a shaded relief effect that enhances visualization.
- Illumination (光照): The brightness of each pixel, which is determined by the angle between the surface orientation and light source direction.
- Shadow (陰影): Areas where light cannot be illuminated due to the cover of the terrain



Viewshed Analysis (視域分析)

- The viewshed analysis tools are useful when you want to know how visible objects might be.
- A viewshed identifies the cells in an input raster that can be seen from one or more observation locations.
- Each cell in the output raster receives a value that indicates how many observer points can be seen from each location.
- Cells can see observer point are given a value of 1, and others are 0.

